Algebra 1
Unit 2
Lesson 1 Practice

Name______________________
Date______________________
Period______________________

1. The power of a power law of exponents states to keep the
- ○ base
- ○ coefficient
- ○ exponent

and then
- ○ add
- ○ subtract
- ○ multiply

the powers.

2. Diana simplified the algebraic expression and made an error. Find and fix the mistake.

$$(-3x^3y^5)^3(1.5xy^0)$$

$$(-3x^9y^{15})(1.5x)$$

$$(-3y \cdot 1.5)(y^{15})(x^9 \cdot x)$$

$$-4.5x^9y^{15}$$

3. Write a note to Diana explaining the error she made in the previous question.

Determine an equivalent algebraic monomial expression for each expression using the laws of exponents. Show your work.

4. $(-8a^5b)(3ab^4)$

5. $(\frac{-1}{4}x^3y^2)(4x^8y^3)(-5y^2)$

6. $(2.8kj^9)(3.5k^8)\left(\frac{1}{3}k^5j^2\right)$

7. $(-3mn^5p^4)^3$

8. $(2.3c^4d)(-2.4c^{-3}d^0)(-c^2d^{-5})$

9. $\left(\frac{5}{6}r^{-8}\right)^{-2}(-9.6r^5s^4)^0$

10. $(-8y^5z)^3(5y^7z^6)^3$

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Find the quotient of each expression by using the laws of exponents to generate an equivalent expression.

1. $\frac{3x^6y^5}{12xy^8}$

2. $\frac{2a^2b^7}{12a^3b^4}$

3. $\left(\frac{-2s^3t^6}{8t^6}\right)^5$

4. $\frac{-2k^5j^9}{-6k^4j^{10}}$

5. $\frac{28w^8z^6}{42w^{10}z^4}$

6. $\left(\frac{xy^9}{-5y^4}\right)^0$

7. $\frac{6r^8s^5}{-12r^5s^8}$

8. $\frac{5x^5}{60x^{11}s^3}$

9. $\left(\frac{13x^0z^5}{-x^4z}\right)^2$

10. $\left(\frac{-25r^{10}p^8}{-10r^4z}\right)^3$

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Lesson 3 Practice | Period____________________

1. Pam was simplifying the expression and made an error. Her steps are shown. Find and fix the mistake.

$$\left(\frac{4x^3y^3 \cdot 2xy}{3xy^2}\right)^2$$

Step 1: $\left(\frac{8x^4y^4}{3xy^2}\right)^2$

Step 2: $\frac{64x^6y^6}{9x^2y^4}$

Step 3: $\frac{64x^4y^2}{9}$

2. Write a note to Pam explaining her error.

For questions 3 through 5, find an equivalent expression using the laws of exponents.

3. $\frac{14x^6y^6}{12x^{10}y^3}$

4. $\frac{4x^6y^6}{6x^2} \cdot \frac{3x}{4y}$

5. $\left(\frac{4x^2y^3}{5x^0y^5}\right)^2$

6. Prove $2^{(2x+4)} = 16 \cdot 2^{(2x)}$ by using the laws of exponents.

7. Prove $4^{(x-2)} = \frac{4^x}{16}$ by using the laws of exponents.

8. Write an equivalent expression for $(-3.1)^{(4x)} \cdot (-3.1)^{(2x+7)}$

9. Use the law of exponents to write an equivalent expression for $5^{(7-3x)}$.

10. Use the law of exponents to write an equivalent expression for $(2.8)^{(6x+8)}$.

Algebra 1
Unit 2
Lesson 4 Practice

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1. True or False? *The polynomial* $5xy^7 + 3x^4y^2 - x^2y^2$ *is written in standard form.*

2. Write the polynomial $-10xy^5 + 2x^3y^2 - 6x^2y^2$ in standard form.

For questions 3-4, rewrite each polynomial in standard form, then identify the features of the polynomial.

3. $4y^2 - 8y^3 + 2 - 6y$

Standard Form ________________

Number of Terms ________

Degree of the polynomial ________

Leading Term ________

Leading Coefficient ________

4. $-3mp^2 + 7m^2p^2 - 4p - 6m^4p^2$

Standard Form ________________

Number of Terms ________

Degree of the polynomial ________

Leading Term ________

Leading Coefficient ________

5. Create a polynomial that has 3 terms, a degree of 6, a leading coefficient of 5, and a constant term of -1.8.

6. Alexis found the product of $3x^2(5x^4 - 3x^3 + 7x - 8)$ to be $15x^8 - 9x^6 + 21x^3 - 8x^2$. Is she correct? Explain.

7. Find the product of $-2r^2(12r^3 - 2r^2 - 9r + 11)$ in standard form.

Find the product of each by distributing the monomial to the polynomial. Express the product in standard form.

8. $\frac{1}{2}y^2(6y^2 - 6y + 7)$

9. $(6xy^2 + 4x^2y + 7xy + 8)\left(\frac{2}{3}x^3y^2\right)$

10. $-2.8a^3(1.2a^4 - 2.3a^3 + 0.6a)$

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Lesson 5 Practice Period____________________

1. Are the terms $6ab^3$ and $-2a^3b$ like terms? Why or why not? Explain.

2. Find the sum of $(8 - 3t - 5t^2)$ and $(8t + 3t^2)$.

3. Write the sum from question 2 in standard form.

4. Find the difference of $(7.8n^2 + 2.5 - 2n)$ and $(-2n^2 - 8)$.

5. Write the difference from question 4 in standard form.

6. What polynomial must be added to $-3c^2 + 3 - 6c$ in order to get a sum of $2c^2 - c + 5$?

For questions 7 through 10, find each sum or difference. Write the result in standard form.

7. $(5g^2h - 3.2gh^2 + 6) + (-2.75gh^2 + 10)$

8. $\frac{1}{2}r^6t^2 - (2r^3t^2 - r^6t^2 + 3rt^2)$

9. $(18a^2 + 5a - 9b^2) + (3a + 6b^2 - 7a^2)$

10. $(2.7m^2 - 3.1n) - (4n^2 + 5.1m)$

Algebra 1
Unit 2
Lesson 6 Practice

Name____________________
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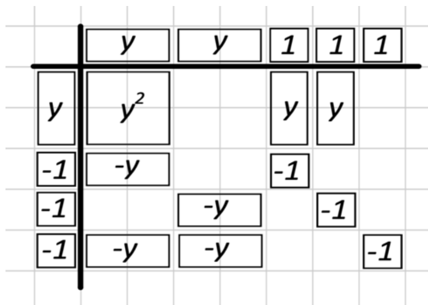
1. Model the following expression with algebra tiles.

$$3x + 5$$

Use the expression $(2y + 3)(y - 3)$ for questions 2 through 4.

2. Complete the model to find the product of $(2y + 3)(y - 3)$ by drawing and labeling the missing tiles.



3. Determine how many of each tile are represented in the product and write the numbers in the table.

y^2 **tiles**	$+y$ **tiles**	$-y$ **tiles**	$+1$ **tiles**	-1 **tiles**

4. Complete the statement.

The product of $(2y + 3)(y - 3)$ is ____________________.

For questions 5-9, use algebra tiles to find each product.

5. $(2 - 5x)(x - 2)$ = ____________________

6. $(3x+1)^2 =$ ______________________

7. $(x+4)(x-4) =$ ______________________

8. $(x-3)(x+2) =$ ______________________

9. $(2-x)(3x+1) =$ ______________________

10. One of your classmates was absent when you learned how to multiply polynomials using algebra tiles. Describe to your classmate the most important thing to remember when multiplying with algebra tiles.

Algebra 1
Unit 2
Lesson 7 Practice

Name____________________
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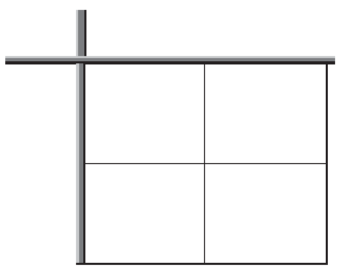
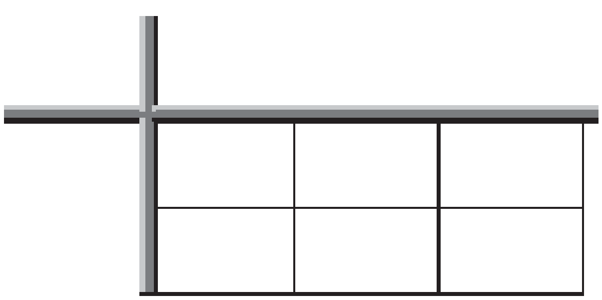
1. Find the product of $(x^2 + 4)(2x - 5)$ using the distributive property.

2. Find the product of $(x^2 + 2)(3x + 4)$ using an area model.



3. Find the product of $\left(-\frac{1}{2}y + 4y^2 - 3\right)(4y - 5)$ using the distributive property.

4. Find the product of $\left(\frac{1}{3}y + 2y^2 - 6\right)(y - 4)$ using an area model.



Find each product.

5. $\left(-4a+\frac{3}{5}b\right)^2$

6. $(2.2\ -5c)(0.7+4c^3-c^2)$

7. $\left(\frac{3}{5}x-2\right)\left(\frac{-1}{2}+4x\right)$

8. $\left(-4\frac{1}{2}+4d^2\right)(5d-7.1c-2d^2)$

9. $(14+4m)\left(-6n-\frac{14}{5}\right)$

10. $\left(3r^2-\frac{5}{4}+15r\right)(-2r+3r^2+12)$

Algebra 1 Name____________________
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Lesson 8 Practice Period____________________

1. Rewrite the expression in each step using the description: $6(x+4)+3x$

	Distribute the 6 to $(x+4)$.
	Combine like terms.

2. Rewrite the expression in each step using the description: $-3x+5(-2x-5)$

	Distribute the 5 to $(-2x-5)$.
	Combine like terms.

Use the order of operations to simplify each polynomial.

3. $3(x+5)+2(x-8)$

4. $\frac{3}{5}(x-6)+3x$

5. $-2(x^2 + 4x) + 7(x - 2)$

6. $2x^2 - 45(x - 7)$

7. $(x + 3)(x - 5) - 9(2x^2 + 3x)$

8. $-\frac{1}{3}(6x - 12) + 7x$

9. $2x^2 + (x + 5)(x - 5)$

10. $3\mathrm{x}(\mathrm{x} - 4) + 7(\mathrm{x} - 3)$

Algebra 1
Unit 2
Lesson 9 Practice

Name____________________
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1. Use the area model to find $(2.4y^3 + 1.4x^3y^2 - 3.2x^2y) \div (2xy)$ where $x \neq 0$ and $y \neq 0$.

Find each quotient.

2. $\left(\frac{1}{4}x^3 + \frac{1}{3}x^2\right) \div \left(-\frac{1}{4}x^2\right)$ where $x \neq 0$

3. $(5a^3 + 10a^2 - 30a) \div (5a)$ where $a \neq 0$

4. $(6k^6r^5 + 3k^3r^4 - 12k^3r^2) \div (3k^2r)$ where $k \neq 0$ and $r \neq 0$

5. $(-24w^6 + 12w^3 - 16w^2) \div (4w)$ where $w \neq 0$

6. $\frac{14x^4-42x^3+21x^2-28}{7x}$ where $x \neq 0$

7. $\frac{12p^4+42p^3+30p^2-120p}{6p}$ where $p \neq 0$

8. $\frac{4t^4-8t^3+12t^2-8t}{4t}$ where $t \neq 0$

9. $\frac{3.5x^3+4.5+3.5x^2-0.5}{0.5x}$ where $x \neq 0$

10. $\frac{16y^3+46y^2+40y^2}{-4y^3}$ where $y \neq 0$